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Topic: Linux Sudo Configuration and Privilege Management

1. Create a new user account called zomatoadmin on your Linux system, and configure it so that this user can run only the apt-get update and apt-get upgrade commands using sudo without entering a password.

Hint: Edit the /etc/sudoers file using visudo and use the NOPASSWD directive for specific commands.

⇒ Step 1: Create the user

```
sudo useradd -m zomatoadmin
```

```
sudo passwd zomatoadmin
```

Step 2: Edit the sudoers file

```
sudo visudo
```

Add the following line:

```
zomatoadmin ALL=(ALL) NOPASSWD: /usr/bin/apt-get update,  
/usr/bin/apt-get upgrade
```

Step 3: Test

Switch to the user:

```
su – zomatoadmin
```

Run:

```
sudo apt-get update
```

```
sudo apt-get upgrade
```

2. Modify the sudo privileges for your existing user so that they cannot run sudo rm -rf / but can run all other sudo commands.

Constraint: Use command-specific restrictions in the sudoers file to block only the dangerous command.

⇒ Open the sudoers file:

```
sudo visudo
```

Assume your username is student.

Add:

Cmnd_Alias DANGEROUS = /usr/bin/rm -rf /

student ALL=(ALL) ALL, !DANGEROUS

Test

Allowed:

sudo apt update

sudo mkdir /tmp/test

Blocked:

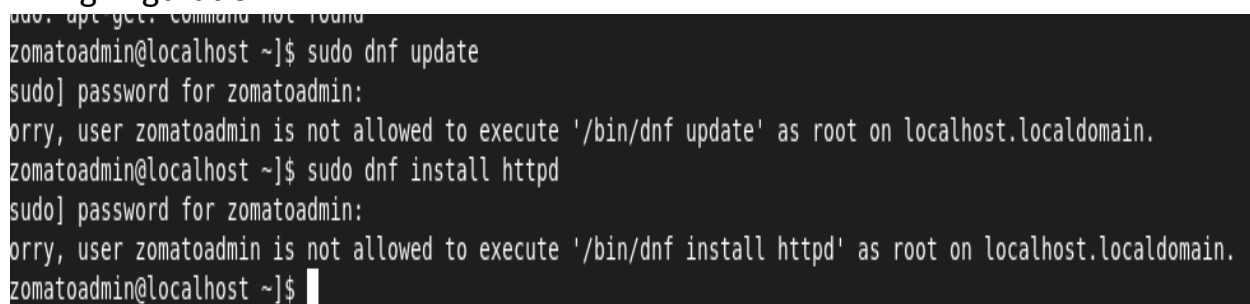
sudo rm -rf /

3. Simulate a scenario where a user tries to run a command with sudo that they are not allowed to. Take a screenshot of the error message and explain in 2-3 lines why the error occurs based on your sudoers configuration

⇒ Commands

sudo apt-get install

nginxguration.



```
zomatoadmin@localhost ~]$ sudo dnf update
sudo] password for zomatoadmin:
orry, user zomatoadmin is not allowed to execute '/bin/dnf update' as root on localhost.localdomain.
zomatoadmin@localhost ~]$ sudo dnf install httpd
sudo] password for zomatoadmin:
orry, user zomatoadmin is not allowed to execute '/bin/dnf install httpd' as root on localhost.localdomain.
zomatoadmin@localhost ~]$
```

4. Set up a group called myntragroup and add two users to it. Grant sudo privileges to this group so that all members can restart the networking service (systemctl restart networking) but cannot edit any files in /etc.

⇒ Create the group

sudo groupadd myntragroup

Step 2: Create two users

sudo useradd user1

sudo passwd user1

```
sudo useradd user2
```

```
sudo passwd user2
```

Step 3: Add users to the group

```
sudo usermod -aG myntragroup user1
```

```
sudo usermod -aG myntragroup user2
```

Verify:

```
id user1
```

Step 4: Find the correct service name

RHEL 9 usually uses NetworkManager.

Check:

```
systemctl list-units --type=service | grep -i network
```

You'll usually see:

NetworkManager.service

Step 5: Edit sudoers

```
sudo visudo
```

tep 6: Test

Allowed:

```
sudo systemctl restart NetworkManager
```

Not allowed:

```
sudo vi /etc/passwd
```

5. Use ChatGPT to generate a sample `/etc/sudoers` entry that allows a user called `movieadmin` to run all commands as root except for shutting down or rebooting the system. Copy the AI's suggestion, test it on your system, and write one line about whether it worked as expected.

⇒ ChatGPT Suggestion

```
Cmnd_Alias SHUTDOWN_CMDS = /usr/sbin/shutdown, /usr/sbin/reboot
```

```
movieadmin ALL=(ALL) ALL, !SHUTDOWN_CMDS
```

On some distributions, verify the command paths with

which shutdown

which reboot

Test

Allowed:

```
sudo mkdir /tmp/movie
```

```
sudo useradd demo
```

Blocked:

```
sudo reboot
```

```
sudo shutdown now
```

One-line observation:

The configuration worked as expected: movieadmin could run other administrative commands but was denied access to reboot and shutdown commands.